Aszadurrahman Rakin

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Research Interests

Systems Reliabilty for Cloud and AI Systems

Education

Boston University — Boston, MA, USA
Ph.D. in Computer Engineering — *September 2025 – Present*

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering; ***CGPA: 3.89/4.00*** — *April 2019 – July 2024*

Work Experience

Bangladesh University of Engineering and Technology — Dhaka, Bangladesh
Research Assistant, Dept. of CSE — *July 2024 – August 2025, Full-time*

- Automated Software Testing

Research Experience

Silent Performance Issue Detection in Cloud Systems — Research Project
Boston University — *September 2025 – Present*

- Investigated silent performance issues open source cloud systems.
- Developed a prototype tool for silent performance issue detection.

Power Aware Scheduling of AI Workloads — Research Project
Boston University & University of Washington — *April 2026 – Present*

- Studied workloads that monopolize GPU resources, and the subsequent GPU behavior.

Symbolic Execution to Find Bugs in Update Patches — Research Project
University of Washington and Vanderbilt University — *September 2024 – February 2025*

- Gained proficiency with symbolic execution engine techniques
- Investigated program slicing methods to identify optimal sets of variables for symbolic analysis

Automated Android Testing Framework — Research Project
Bangladesh University of Engineering and Technology & Samsung Research Bangladesh — *June 2024 – December 2024*

- Contributed to the key algorithm/interaction mechanisms, specifically focusing on handling dynamic cycles in automated GUI interaction.
- Engineered test case generation strategies to improve bug detection and system failure identification

Extracurriculars

Competitive Programming

- **ICPC Regionals Dhaka Site**
- **CodeForces**: Max Rating: 1652

Skills

Programming Languages: Go, C/C++, Java, Python, JavaScript, TypeScript, SQL, XML, x86 Assembly, MIPS, LaTeX, Bash

Operating Systems: xv6, Linux, Windows

Compiler Design Tools: Flex, Bison

Microcontrollers/Embedded Systems: Arduino, ATmega32

Framework and Libraries: Appium, ReactJS, expressJS, FastAPI, NextJS, PyTorch, HuggingFace, NLTK, Langchain

Database Management: OracleDB, PostgreSQL, MongoDB, Redis

Version Control System: Git

Cloud: AWS(VM), Google Cloud Platform(Maps), Azure(Voice Recognition, VM)